



Assignment 12

Part1 (social media): Apply everything the instructor explained in the session for social media app (4 Grades)

Part2 (Chat App): Using mongoose defines the following collections (6 Grades):

User	Message	Room
• name (String, required) • email (String, Unique, required) • Password (String, required)	• roomId (ref Room, required) • senderId (ref to User, required) • content (String, required)	• users (array of user) • name (String, unique, required) • createdBy (ref user, required)

A- REST APIs (3 Grades)

1. Signup. (0.5 Grade)

- URL: POST /auth/signup

```
{
  "name": "user1",
  "email": "user1@email.com",
  "password": "password"
}
```

```
{
  "message": "User registered successfully"
}
```

2. Login user and return JWT token. (0.5 Grade)

- URL: POST /auth/login

```
{
  "email": "user@email.com",
  "password": "password"
}
```

```
{
  "message": "Login successful",
  "token": "jwt_token_here"
}
```

3. Create a new chat room. (0.5 Grade)

- URL: POST /room

```
{
  "name": "TechTalk"
}
```

```
{
  "message": "Room created successfully",
  "room": {
    "id": "65f4a2c5e4b0d6a3f7e9b123",
    "name": "TechTalk"
  }
}
```

4. Get all available rooms. (0.5 Grade)

- URL: GET /room

```
"rooms": [
  { "id": "65f4a2c5e4b0d6a3f7e9b123", "name": "TechTalk" },
  { "id": "65f4a2c5e4b0d6a3f7e9b115", "name": "GamingChat" }
]
```

5. Delete a Room (By Owner Only). (0.5 Grade)

- URL: DELETE /room/:roomId

```
{
  "message": "Room deleted successfully"
}
```

```
{
  "error": "Access denied. Only Owner can delete rooms."
}
```

6. Get all chats for specific room. (0.5 Grade)

- URL: GET /room/:roomId

```
{
  "messages": [
    {
      "sender": "User1",
      "content": "Hello everyone!",
      "timestamp": "2025-02-08T10:00:00Z"
    },
    {
      "sender": "User2",
      "content": "Welcome to TechTalk!",
      "timestamp": "2025-02-08T10:02:00Z"
    }
  ]
}
```



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B- Socket Events (3 Grades)

7. Join a room. (1 Grade)

- Event: "joinRoom"
- Input:

```
{  
  "roomId": "65f4a2c5e4b0d6a3f7e9b123"  
}
```

- Output (Broadcast to all users in room):

```
{  
  "message": "User1 has joined the chat"  
}
```

8. Send a message to a specific room. (1 Grade)

- Event: "sendMessage"
- Input:

```
{  
  "roomId": "65f4a2c5e4b0d6a3f7e9b123",  
  "content": "Hello everyone!"  
}
```

- Output (Broadcast to all users in room):

```
{  
  "sender": "User1",  
  "content": "Hello everyone!"  
}
```

9. Leave a room. (1 Grade)

- Event: "leaveRoom"
- Input:

```
{  
  "roomId": "65f4a2c5e4b0d6a3f7e9b123"  
}
```

- Output (Broadcast to all users in room):

```
{  
  "message": "User1 has left the chat"  
}
```

⚠ Important Notes about postman

1. Name the endpoint with a meaningful name like 'Add User', not dummy names.
2. Save your changes on each request (ctrl+s).
3. Include the Postman collection link (export your Postman collection) in the email with your assignment link

Bonus (3 Grades)

How to deliver the bonus?

- 1- Solve the problem [Rotate Array - LeetCode](#) on **LeetCode**
- 2- Inside your assignment folder, create a **SEPARATE FILE** and name it "bonus.js"
- 3- Copy the code that you have submitted on the website inside "bonus.js" file